

Asteroid Photometry

Joan Ribot Oliver

Master in Particle Physics and Cosmology, University of Cantabria

10th June 2026

Abstract

The main goal of the project is to do a photometric study of a known asteroid to obtain its light curve and rotation period (intentionally chosen to be short, about 10-12h), which still remain unknown/not published. The data reduction is performed using the FotoDif software by Julio Castellano [1], which allows the analysis of the data stored in separated .fit files. These contain the main data (8271 Imai), the dark field (darks), the flat field (flat) and the bias (bias) corrections, respectively. Last but not least, a second objective of the project is to work with the main telescope of the Observatorio Astronómico de Cantabria (OAC) and to learn the procedure and workflow of a real observation session in May 2026.

1 Introduction

The Solar System is a vast and complex system that consists of the Sun, planets, moons, asteroids, comets, and other small bodies. In this project, we focus on the study of asteroids, which are small rocky bodies that orbit the Sun, primarily found in the asteroid belt between Mars and Jupiter. These are remnants from the early solar system, providing valuable insights into its formation and evolution. To study asteroids, astronomers often use photometry, which involves measuring the brightness of an asteroid over time and produces the light curve. This information allows to determine the asteroid's rotation period, and some properties of its shape and surface.

The main goal of this project is to do a photometric study of a known asteroid, named 8271 Imai, to obtain its light curve and rotation period, which still remain unknown or not published. A second but not less important objective of the project is to work with the main telescope of the Observatorio Astronómico de Cantabria (OAC) and to learn the procedure and workflow of a real observation.

2 Observational Data

Before the data gathering, there is a previous selection process to decide which asteroid to observe. This is not a simple task as many factors must be taken into account, such as the visibility of the asteroid from the observatory, its brightness, its expected rotation period, the availability of the telescope, the weather conditions for future observations... a list that never ends.

Data from the Minor Planet Center (MPC, [2]) and the Asteroid Lightcurve Data Exchange Format database (ALCDEF, [3]) are consulted to check the known information about the asteroid. The 8271 Imai asteroid is selected for this project after my visit to the OAC by the supervisor, Javier Ruiz. The data of this asteroid analysed and shown in this project are from observation sessions for May 15, 22, 23, June 05 and 07, 2026. There is no data available for the weekend of May 29 and 30 because of weather conditions and full moon. During my visit on May 8, we have discarded the 16583 Oersted asteroid as no magni-

tude variation is seen after 30 minutes of observation.

The procedure to gather the data itself is the following: first, the 40 cm aperture diameter telescope of the OAC is turned on and the necessary calibrations are performed, such as the cooling of the CCD with a thermocouple and the focus of the CCD camera. Secondly, the asteroid is located using its coordinates and the TheSkyX software, and the telescope is pointed towards it. Then, a reference star close to the asteroid, so they are in the same field of view (FoV), is selected so that the telescope can be guided during the observation session. Finally, an image of the field with the asteroid in the center is taken every minute for a prudential time to check if there is any magnitude variation, such as in the previously mentioned case of the 16583 Oersted asteroid. If there is no magnitude variation, the asteroid is discarded and another one is selected. If there is a magnitude variation, the observation session continues for a longer time, such as 4-5 hours, to collect enough data to obtain the light curve of the asteroid for that night. The procedure is repeated for several nights to gather more data so the period of rotation can be calculated with more precision.

During the night there might be some problems, such as the weather conditions, the tracking of the telescope and the dome, the focus of the images, the transit of the asteroid near bright stars, the presence of the Moon, etc. In my case, the main problem is the weather conditions: at about 3:00 am the sky gets completely covered by fog, which forces us to stop with the observation session. However, we are able to discard the 16583 Oersted asteroid. I have not been able to stay for more nights at the observatory because of logistics and legal factors.

3 Methodology and Data Analysis

To process the data I use the FotoDif software (.exe) by Julio Castellano [1]. In my particular case, I need to use a virtual machine (VM) with a Microsoft Windows OS as I own a MacOS. Before the observation session at the OAC, I have been provided by Javier Ruiz with some mock data from previous observation sessions at OAC of the asteroids 1818 Brahms and 6147 Straub, which I have used

to practice the data analysis pipeline and to learn how to use the software. All data have the same structure of separated .fit files: the main data (8271 Imai, 1818 Brahms, 6147 Straub), the dark field (darks), the flat field (flat) and the bias (bias) corrections, respectively.

This is the workflow folloed: first, FotoDif is used to get the coordinates of the asteroid and to perform the photometric analysis of the images. Then, Aladin Lite [4] is used to find out which one is the asteroid (VAR-1) and to know the absolute magnitude (M) of the calibration stars. Finally, the Julio Castellano [1] tutorial is followed to get the light curve and the rotation period of the asteroid.

4 Results and Discussion

The main results of the project are the light curve and the period of rotation of the 8271 Imai asteroid. The first can be seen in Fig. 1, where the absolute magnitude variation of the asteroid is plotted against the time of observation.

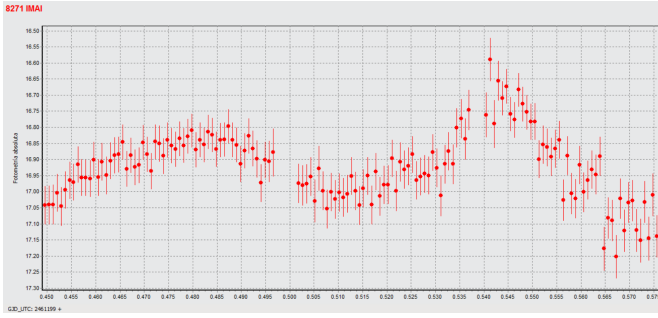


Fig. 1: Absolute Magnitude variation against time of 8271 Imai during the observation session of June 7, 2026.

This light curve almost completes a full period of rotation of the asteroid and it shows a magnitude variation of about 0.8 mag. It presents two maximums and two minimums, which means that it is dominated by the shape of the asteroid. Nevertheless, there is also presence of albedo features on the asteroid surface that reflect light differently as it rotates, as the two different maximums and minimums are not symmetric.

FotoDif [1] allows to get other plots such as for the Air-mass, the Transparency, the FWHM, etc. However, they are not shown here for the sake of brevity and because they are not the main result of the project.

After processing the data of the different observation sessions, the light curves are stacked to get a better estimate of the period of rotation, shown in Fig. 2. This plot shows the variation of the relative magnitude of the asteroid against the phase of the period of rotation.

The main data obtained for the period of rotation of 8271 Imai are shown in Tab. 1.

Asteroid	Period (d)	Period (h)	StdDv
8271 Imai	0.13403	3.21672	0.04678

Tab. 1: Main data for the period of rotation of 8271 Imai. The standard deviation is minimized by stacking the light curves of the different observation sessions.

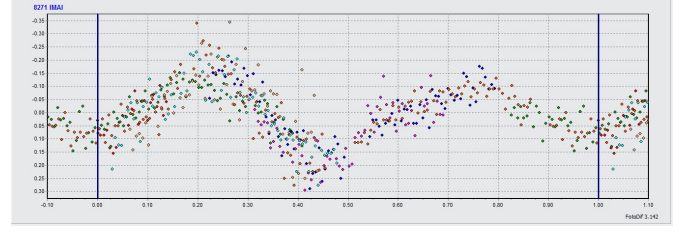


Fig. 2: Stacked relative magnitude light curves against the phase of rotation of 8271 Imai so the standard deviation is minimized

The period of rotation is estimated to be about 0.134 days around its axis, which is about 3.2 hours. This is a short period of rotation, so the asteroid is a fast spinner. The standard deviation of the data points is about 0.047 mag, a good result considering the magnitude variation of the asteroid is about 0.8 mag around 17 mag, which is a faint object.

5 Conclusions

In this project, we have performed a photometric study of the 8271 Imai asteroid to obtain its light curve and rotation period, which result to be about 0.8 mag and 0.134 days (about 3.2 hours), respectively. The light curve is unique: it presents two maximums and two minimums, which means that it is dominated by the shape of the asteroid, but it also shows presence of albedo features on its surface that reflect light differently as the asteroid rotates, as the these maxima and minima are not symmetric.

The main challenges faced during the project were related to the weather conditions, which forced us to stop the observation session at about 3:00 am due to the presence of fog, and to the use of a virtual machine to process the data, which caused some problems with the plots and the output of the software.

The results obtained for this asteroid, altogether with many other asteroids, will be published in the trimestral report of the Minor Planet Bulletin (MPB, [2]) after a more detailed analysis of the data and a more precise estimation of the period of rotation is done.

References

- [1] Rodeno, J. (2026). *FotoDif: Software para fotometría digital diferencial de estrellas variables y asteroides*. <http://www.astrosurf.com/orodeno/fotodif/index.htm>
- [2] Minor Planet Center. (2026). *The Minor Planet Bulletin*. Association of Lunar and Planetary Observers (ALPO). <https://mpbulletin.org/>
- [3] Warner, B. D. (2026). *Asteroid Lightcurve Data Exchange Format (ALCDEF) Database*. https://alcdef.org/php/alcdef_GenerateALCDEFPage.php
- [4] Boch, T., & Fernique, P. (2014). *Aladin Lite: A planetarium in your browser*. Astrophysics Source Code Library, record ascl:1412.008. <https://aladin.cds.unistra.fr/AladinLite/>